Lorenzo Mascia

17/12/2001

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Work Experience

Digital Education Content Creator Feb. 2026 – Apr. 2026
University of Sassari – DEH ALMA Project *Remote*

- Designed and developed an **Open Lesson for a Fundamentals of Computer Science course** within the DEH ALMA digital education project.
- Produced structured **digital learning materials** and educational audio/video content for online and self-paced learning.

Research Assistant Nov. 2023 – Sep. 2024
University of Sassari *Sassari, Italy*

- Developed and evaluated point-cloud compression methods to improve the efficiency of **real-time transmission** of large-scale 3D data.
- Worked with **Intel RealSense RGB-D cameras** and SDK for image and depth acquisition, integrating sensor data into computer vision pipelines.
- Developed a **mixed-reality metaverse platform in Unity** enabling multiple users to communicate and interact with shared 3D data.
- Integrated concepts related to **Human Digital Twins (HDT)** and **Self-Sovereign Identity (SSI)** into the mixed-reality platform.

Education

Politecnico di Torino Turin, Italy
M.Sc. in Computer Engineering, Graphics and Multimedia – GPA 29.3/30 *Sep. 2024 – Expected Dec. 2026*

University of Sassari Sassari, Italy
B.Sc. in Computer Engineering – 110/110 cum laude *Oct. 2023*

Projects

Real-World Guided Depth Super-Resolution | *Master's Thesis, Python, PyTorch, Deep Learning* 2026

- Designed BAF-Net, a novel deep learning architecture integrating RGB guidance and Depth Anything V2 priors for metric depth reconstruction.
- Built preprocessing and evaluation pipelines for real-world RGB-D and LiDAR datasets, addressing sensor alignment and depth quality.
- Achieved state-of-the-art performance on the TOFDSR benchmark with a final RMSE of **4.011 cm**.

SSI Metaverse | *Unity, C#, TypeScript, Veramo, SSI* 2024

- Developed a **multi-user mixed-reality environment in Unity** enabling real-time sharing and interaction with 3D data.
- Integrated a **Self-Sovereign Identity architecture** using Veramo to manage decentralized digital identities in a Human Digital Twin-oriented metaverse environment.

Publications

Saliency-Guided Point Cloud Compression for 3D Live Reconstruction May 2024

- P. Ruiu, **L. Mascia**, E. Grosso. *Multimodal Technologies and Interaction* (MDPI)

Towards immersive communication: A user-centered investigation of life-sized 3D telepresence Sep. 2026

- P. Ruiu, I. Giorgi, F. Casu, L. Mascia et al. Computers in Human Behavior Reports.

Additional Information

Programming Languages: Python, C/C++, C#, Java, Rust, SQL, JavaScript, TypeScript, MATLAB, Verilog
Technologies: OpenCV, PyTorch, TensorFlow, Unity, Blender, Docker, git, bash, Jupyter, React
Languages: English C1, Italian native